Two roots, and that is all

$x^2 + x = 0 \iff x \in \{0, 1\}$ in characteristic 2, read slowly

Abstract

One line of algebra.

- `sq_add_self_eq_zero_char2`: $u^2 + u = 0 \iff u = 0 \vee u = 1$.

It is the literal *last step* of `kasami_is_apn`: once a collision has been squeezed to $(x + y)^2 + (x + y) = 0$, this lemma delivers "at most two solutions", i.e. APN.

Setting

$$\Gamma = \big(F : \text{Type},\ [\text{Field } F],\ [\text{CharP } F\ 2]\big)$$

A field where $1 + 1 = 0$, so $-1 = 1$ and $u + u = 0$ for every u.

1 The lemma

$$\forall u : F. \qquad u^2 + u = 0 \longleftrightarrow (u = 0 \ \vee \ u = 1)$$

```
lemma sq_add_self_eq_zero_char2 {F : Type*} [Field F] [CharP F 2] {u : F} :
    u ^ 2 + u = 0 ↔u = 0 ∨u = 1 := by
  grind +suggestions
```

`grind` closes it because the statement is a polynomial identity over an integral domain: it factors and reads off the roots. We read the factoring by hand below.

Why it is true (the factoring)

Factor the left side:

$$u^2 + u \;=\; u(u+1).$$

In a **field** (no zero divisors) a product is zero iff a factor is zero:

$$u(u+1) = 0 \iff u = 0 \;\vee\; u+1 = 0.$$

In characteristic 2, $u + 1 = 0 \iff u = 1$ (because $-1 = 1$). Hence $u \in \{0, 1\} = \mathbb{F}_2$.

$$u^2 + u = 0 \xleftrightarrow{\text{factor}} u(u+1) = 0 \xleftrightarrow{\text{domain}} u = 0 \;\vee\; u = 1$$

It is an Artin–Schreier polynomial $X^2 - X = X^2 + X$ in char 2. Its roots are exactly the prime subfield $\mathbb{F}_2$. The map $u \mapsto u^2 + u$ is $\mathbb{F}_2$-linear with kernel $\{0, 1\}$ — this is the additive shadow of "$x \mapsto x^2$ is Frobenius".

2 Where it lands: the last line of Kasami

Inside `kasami_is_apn` a collision between x and y is reduced, via the trace identity, to

$$x^2 + x \;=\; y^2 + y.$$

Add the two sides and use Freshman's Dream $(x + y)^2 = x^2 + y^2$:

$$(x+y)^2 + (x+y) \;=\; (x^2 + x) + (y^2 + y) \;=\; 0.$$

Now the lemma fires on $u = x + y$:

$$(x+y)^2 + (x+y) = 0 \xRightarrow{\texttt{sq\_add\_self\_eq\_zero\_char2}} x + y = 0 \;\vee\; x + y = 1.$$

That is exactly $y = x$ or $y = x + 1$ — two solutions, which after rescaling by a is the APN condition.

```
have h_sum_zero : (x + y) ^ 2 + (x + y) = 0 := by
  have h2 : (x + y) ^ 2 = x ^ 2 + y ^ 2 := add_pow_char (R := F) (p := 2) x y
  rw [h2]
  have : x ^ 2 + y ^ 2 + (x + y) = (x ^ 2 + x) + (y ^ 2 + y) := by ring
  rw [this, hu, CharTwo.add_self_eq_zero]
rw [sq_add_self_eq_zero_char2] at h_sum_zero
```

Compare with the Frobenius pair. `frob_bijective`+`frob_add` open the door (transport APN through a twist); this lemma closes it (read off the two roots). Both are tiny; both are decisive.

One-screen summary

step	**content**
factor	$u^2 + u = u(u+1)$
domain	product $= 0 \Rightarrow$ a factor $= 0$
char 2	$u + 1 = 0 \iff u = 1$
land	$(x+y)^2 + (x+y) = 0 \Rightarrow y \in \{x,\, x+1\}$ (APN)

$$u(u+1) = 0 \text{ in a field of char 2 } \Rightarrow \text{ Kasami collision has} \leq 2 \text{ solutions } \Rightarrow \text{ APN.}$$